COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	08-26-2026	**DATE SUBMITTED:**	08-26-2026
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Prelim Laboratory Exam	**Set D**

1. PROCEDURE OUTPUT

What procedures did you perform?

Algorithm

1. Create an array-based Stack.
2. Create a linked list-based Queue.
3. Get the user to enter a sequence of positive integers.
4. Store the input values in the sequence.
5. Iterate through the input sequence. For each value:
6. Push the value onto the Stack.
7. Enqueue the value into the Queue.
8. Initialize a boolean flag *isSymmetric* to true.
9. While the Stack is not empty:
10. Pop the top value from the Stack.
11. Dequeue the front value from the Queue.
12. Compare the two values.
13. If they are not equal, set *isSymmetric* to false and break the loop.
14. If *isSymmetric* is true, display: "The sequence is symmetric."
15. If *isSymmetric* is false, display: "The sequence is not symmetric."

Analysis:

This algorithm basically compare the two values from the values that the user input then it is stored in the function called sequence then for each value we push the value into the stack and enqueue the value to the queue.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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